Empowering Physical Attacks With Jacobian Matrix Regularization Against ViT-Based Detectors in UAV Remote Sensing Images

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***Abstract*— Vision transformers (ViTs) have achieved great success in unmanned aerial vehicle (UAV) target detection tasks. However, little attention has been paid to the adversarial attack against ViT-based detectors, and the generated adversarial examples cannot take physical realizability and attack transferability into account at the same time. To overcome the limitation, we focus on transferable attacks toward ViT-based detectors in optical UAV-based remote sensing images and generate adversarial examples in the physical world. Concretely, we design unique perturbation patches deployed within and beyond the target object rather than requiring the patches to be aligned with image tokens. To narrow the gap between limited digital samples and complex physical scenarios, we conduct data augmentation on training images at global and local levels. In addition, we propose a novel transferable attack method named Jacobian matrix regularization (JMR), which consists of feature variance regularization (FVR) and attention weight regularization (AWR). Specifically, FVR calculates feature variances of different channels within specific layers and then sets the features as zeros for channels with top variances. AWR is achieved by masking the largest self-attention weights. We conduct extensive transferable experiments with typical detectors in both digital and physical UAV-based remote sensing scenarios. The results indicate that our method could achieve competitive transferability compared with state-of-the-art methods.**

***Index Terms*— Jacobian matrix regularization (JMR), object detection, physical attacks, unmanned aerial vehicle (UAV) remote sensing, vision transformers (ViTs).**

I. Introduction

AS ONE of the most fundamental tasks of the remote sensing field, object detection has undergone rapid development since entering the era of deep learning [1]. To accomplish this task, convolutional neural networks (CNNs) [2], [3], [4] have played a pivotal role in feature extraction. However, since their introduction in 2020, vision transformers (ViTs) [5] seem to have taken the spotlight from CNNs. Since then, many variants of ViTs [6], [7] have been proposed to enhance their detection performance in satellite images or aerial images [8]. Unlike convolutional operations constrained by limited receptive fields, ViTs typically transform input images into a sequence of patches and extract the global relationship with the self-attention mechanism, which contributes to capturing long-range contextual information between and within objects. Consequently, ViTs could have equal or better performance than CNNs in various tasks [7]. However, ViTs confront the challenges posed by adversarial perturbations [9]. To enhance the reliability of ViTs in remote sensing applications, it is necessary to uncover their weaknesses by conducting adversarial attacks.

Plenty of methods have been introduced to generate adversarial examples in remote sensing scenarios, predominantly focusing on transfer-based black-box attacks [10], [11]. In contrast to white-box attacks, which rely on model accessibility to generate adversarial examples, transfer-based black-box attacks leverage the transferable properties of adversarial examples to deceive unknown black-box models, thereby posing a significant threat to deep learning-based vision systems. Based on the feasibility of realizing malicious examples in the physical world, adversarial attacks can be categorized as digital and physical attacks. Digital attacks often introduce imperceptible perturbations across entire images. While these attacks excel in digital environments, they are ineffective in the real world as the generated perturbations are undetectable by cameras. Conversely, physical attacks can bridge this gap by producing visible adversarial patterns that can be realized in the physical world. One classic example of such patterns is the 2-D patch depicted in Fig. 1(a). The generated patch can be affixed to or positioned near the targets [12], [13]. For practical considerations, this article focuses on transfer-based physical attacks.

Although there are some transferable attacks on remote sensing, this field is still in its infancy, and three major limitations can be found. First, most adversarial attacks have been executed and verified primarily on CNN-based models, not giving enough attention to how their attack methods perform when applied to ViT-based models in the context of remote sensing. While ViT-based models have been explored in certain attack studies, they tend to disproportionately focus

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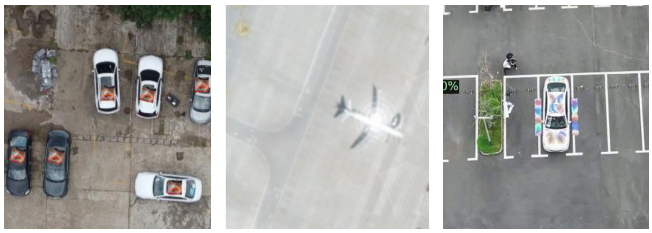
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on attacks between different CNN-based detectors, often limiting their analysis to the Swin Transformer [6] as an example to demonstrate attack transferability [11], [12], [14]. In fact, the various ViT variants exhibit significant differences in their model architectures, and the area of exploring attack transferability between ViT-based detectors remains largely unexplored. Second, many attack methods in remote sensing do not take full account of physical feasibility. Some works prioritize the naturalness of adversarial textures similar to Fig. 1(b), which are challenging to achieve in real-world scenarios [15]. Others focus on imperceptible adversarial patterns that are only effective in digital spaces, ignoring their practical deployability [16]. In addition, some approaches generate adversarial patches by optimizing the background areas surrounding the targets [11], [14]. While this strategy may augment the attack's effectiveness, the patches tend to be significantly larger than the targets, posing difficulties in deploying them in real-world scenarios. Third, despite the existence of patch-based attacks targeting ViT models, these adversarial patches are strictly overlaid on a single or multiple image patch tokens [17]. Even though the yielded patterns can mislead attention modules, the attack performance would deteriorate significantly when the patch is shifted by a certain number of pixels [9].

To overcome the above limitations, we propose a novel method that obviously enhances physical attacks against ViT-based detection models for unmanned aerial vehicle (UAV) remote sensing imagery. First, to strike a balance between the ease of physical deployment and adversarial effectiveness, we devise composite adversarial patches comprising two parts: one deployed within the target object and the other beyond it, as clearly depicted in Fig. 1(c). Second, we adopt a unique data augmentation strategy that encompasses both global and local perspectives, aiming to bridge the gap between digital imagery and complex physical environments. Third, to further enhance attack performance, we conduct an in-depth analysis of self-attention and put forward the Jacobian matrix regularization (JMR) method that incorporates two key components: feature variance regularization (FVR) and attention weight regularization (AWR). FVR assesses the feature variances across different channels within specified layers, subsequently zeroing out the features in channels exhibiting the highest variances. On the other hand, AWR accomplishes its goal by masking the most significant self-attention weights. By employing this approach, we not only mitigate gradient fluctuations but also alleviate the overfitting problem of the self-attention modules, thereby enhancing the transferability of adversarial examples across various models. Finally, we carry out comprehensive transferable experiments utilizing typical detectors in both digital and physical UAV-based remote sensing environments. The results demonstrate that our method can achieve competitive transferability compared with baseline methods.

The contributions of this work are summarized as follows.

1) To balance the convenience of physical implementation and adversarial effects, we design composite perturbation patches deployed both within and beyond the target object. Furthermore, we present a novel data augmentation technique implemented at both the global and local levels, aiming to minimize the disparity between digital images and complex physical scenes.
2) To further enhance attack performance, we propose a JMR method grounded in an in-depth analysis of the self-attention mechanism. This approach comprises FVR and AWR, which erase the features of channels with top variances and maximum self-attention weights, respectively.
3) To verify the effectiveness of our method, we conduct extensive experiments with UAV remote sensing images, which show that our method can achieve competitive transferability compared with typical baseline methods.



(a) (b) (c)

Fig. 1. Typical scenarios of adversarial attacks. (a) Single-patch attacks against UAV images [13]. (b) Natural perturbations against satellite images [15]. (c) Our composite patch attacks against UAV images.

II. Related Works

This section briefly reviews the development of adversarial attacks in remote sensing and transferable attacks on CNN-based models and ViT-based models.

A. Adversarial Attacks in Remote Sensing

Adversarial attacks have been widely studied in the remote sensing community, especially in the attacking scene classification models [16], [18], [19], [20] and the aerial object detection models [12], [14], [15]. A common trait in scene classification attacks is that most experiments solely focused on CNN-based models, neglecting mainstream ViT-based models. Moreover, these studies often created imperceptible adversarial perturbations that could fool deep learning models yet remain undetectable to humans [16], [18], [20]. Consequently, these samples are difficult to realize in real-world scenarios. Regarding attacks on aerial object detection models, several studies have incorporated physical realization into their work. For instance, an adaptive-patch-based physical attack (AP-PA) framework was proposed to mitigate the influence of complex physical factors [12]. This method focused on creating adversarial patches that adapted to physical dynamics and varying scales. This research group also proposed contextual background attack [14] to deceive aerial detectors. Their approach employed the targets as masks for adversarial patches, and only the pixels beyond the masks underwent optimization to guarantee that the crafted patches covered the crucial contextual background. However, the patches are often considerably larger than the targets, thereby presenting challenges in their practical deployment. Beyond patch-based attacks, natural weather-inspired perturbations were proposed

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to deceive optical aerial detectors [15]. Nevertheless, these crafted adversarial patterns also encounter the challenge of deploying into real-world scenarios.

B. Transferable Attacks on CNN-Based Models

Since the advent of adversarial examples [21], researchers have extensively delved into transferable attacks targeting CNN-based models, with a pivotal challenge centered on enhancing the transferability of adversarial samples between different models. One classical strategy is to alleviate the overfitting problem of adversarial samples by applying different data augmentation methods, such as the feature inducing method [22], the remix method [23], and the data-erosion method [24]. Another technical line holds the view that different CNN models have similar intermediate features and attention maps, so the attack on those features or maps could improve cross-model transferability. For instance, one study generated transferable adversarial examples by reducing the similarity between intermediate features of clean and adversarial samples [25]. Another research explored the role of feature attention in enhancing transferability. It was found that modulating the attention ratio between the original and target classes can influence the effectiveness of adversarial examples on unseen models [26]. More recently, a novel approach [10] proposed adjusting the attention distribution using a mask that separates foreground and background attention. This technique allowed for more precise control over the adversarial perturbations, resulting in adversarial samples with improved transferability. Although those methods have shown promising results in attacks against CNN-based models, their application to attacks toward ViT-based models is challenging due to the inherent differences in feature extraction mechanisms. Beyond the aforementioned techniques, scaling backward gradients offers an effective strategy for generating transferable adversarial examples, particularly for models with skip connections. A notable example is the skip gradient method (SGM) [27], which introduced a decay factor to modulate the gradients within residual modules. Based on SGM, learning to learn transferable attack (LLTA) [28] introduced a series of learnable decay factors for residual branches and randomly changed them to achieve model augmentation. Besides, LLTA attributed the effectiveness of decay factors to the explanation that reducing gradients could promote linearity and transferability for Lipschitz continuous models. Although the explanation could make sense for CNNs, it cannot be applied directly to ViTs because self-attention is not Lipschitz continuous. Nevertheless, the above explanation inspires us to propose a method of generating transferable adversarial samples for ViTs.

C. Transferable Attacks on ViT-Based Models

Inspired by transformer architecture in natural language processing [29], ViTs were introduced by splitting images into a sequence of patches before being fed into one standard transformer encoder [5]. Despite the introduction of numerous ViT variants for enhancing their performance, these models are still vulnerable to adversarial examples [9]. In fact, most transferable attacks targeting ViT-based models have been conducted by leveraging their specific architectural features, including the class token and self-attention mechanism. Naseer et al. [30] enhanced the transferability of ViT-based classifiers with the self-ensemble strategy that connected the class token at each layer to a shared head. Besides, they also adopted a token-refinement module to further enhance attack performance. However, their method is difficult to extend to detection tasks due to the absence of class tokens. AttentionFool [31] observed that the gradients of attention modules were much smaller than value tokens, limiting the potential for gradient-based attacks. Thus, it designed adversarial losses with presoftmax dot-product similarities to maximize the number of queries focused on the targeted key, i.e., adversarial patch. Although the method could have promising performance for fixed-location targeted keys, it would be ineffective for dynamic-location or unaligned adversarial patches. PatchFool [17] deceived the self-attention mechanism by designing an attention-aware loss given the selected adversarial patch, but the method could not be applied to situations where adversarial patches were misaligned with image patch tokens. PNA [32] showed that the discard of attention gradients during back-propagation contributed to improved attack transferability, but our experiments show that this approach occasionally leads to counterproductive outcomes in certain scenarios.

III. Proposed Framework

In this section, we introduce a method for crafting physical adversarial patches that attack ViT-based detectors in UAV remote sensing images. The overall framework is outlined in Fig. 2, and the entire process comprises three essential steps. First, we devise a unique perturbation mesh and subsequently generate perturbed images (see Section III-A). Second, we employ both global and local data augmentation techniques to promote transferability (see Section III-B). Finally, to further enhance attack performance, we employ JMR, encompassing FVR and AWR (see Section III-C).

A. Problem Formulation

In this article, we pay attention to generating transferable adversarial samples with ViT-based detection models without requiring perturbation zones to be strictly aligned with image patch tokens. To this end, we draw lessons from the previous 3-D attack method [10] and design a special perturbation mesh $(\mathbf{M}, \mathbf{T})$ around the given 3-D object model $(\mathbf{M}_0, \mathbf{T}_0)$, where $\mathbf{M}_0$ represents the object mesh, $\mathbf{M}$ denotes the perturbation mesh, and $\mathbf{T}_0$ and $\mathbf{T}$ represent the corresponding textures, respectively.

Unlike previous works that select perturbation regions on a given object mesh, we design separate meshes to be optimized based on common contours and then attach them to the surface of $\mathbf{M}_0$. Since ViTs have a strong ability to model global features, the only perturbation within the object would have weak or moderate attack performance, so we extend the perturbation region beyond the object. The process of designing perturbation meshes is displayed in Fig. 2(a), where (a1) denotes a prepared butterfly contour, which could be replaced by any other contours, (a2) is the mesh generated based on (a1), which can be accomplished by common 3-D modeling software such

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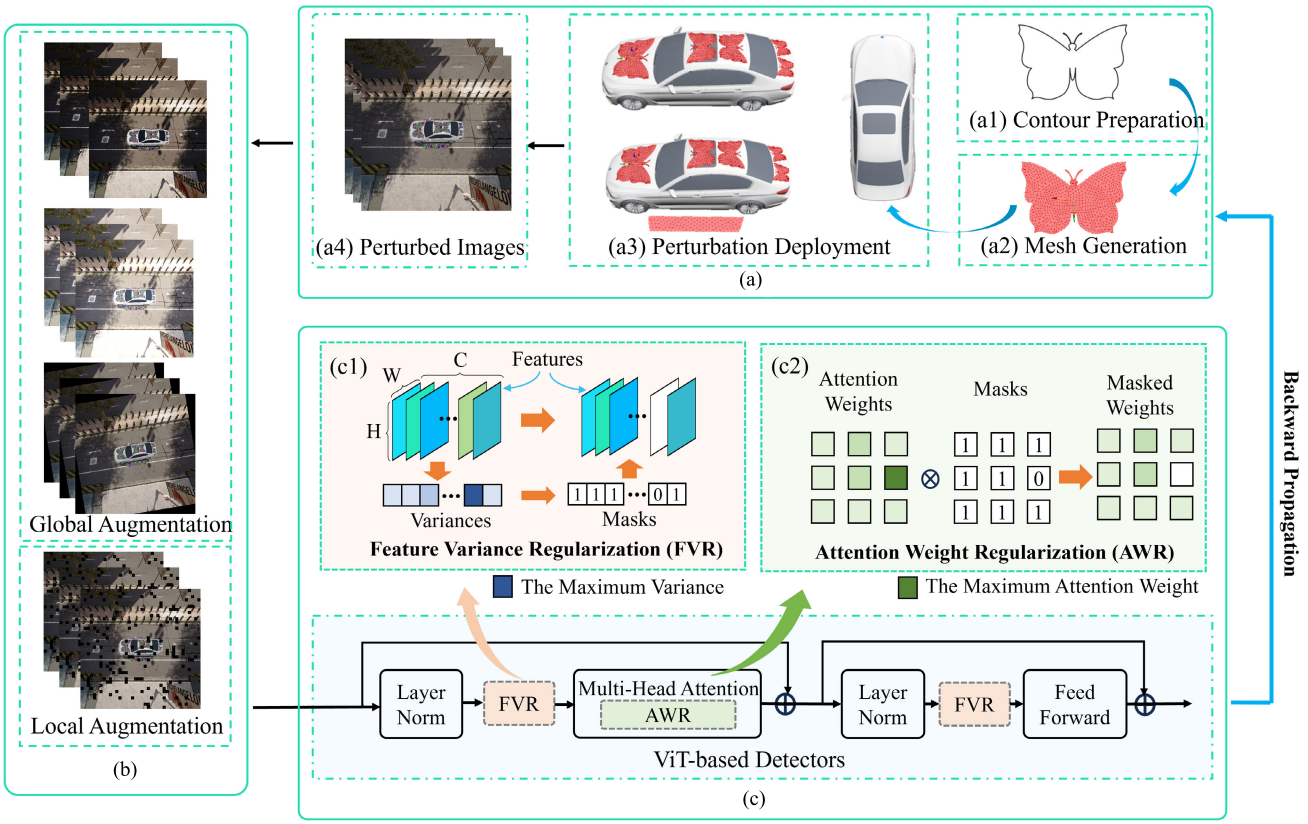


Fig. 2. Pipeline of our proposed method. It mainly includes (a) perturbed images' generation, (b) data augmentation, and (c) JMR.

as Blender, and (a3) shows the object model deployed with perturbation meshes. It should be noted that our experiments explore two types of deployment methods, deploying perturbation meshes within the object and extending perturbations with two additional meshes symmetrically distributed around the object, which could be seen in Fig. 2(a3). We find that the second deployment method could enhance attack performance, which can be seen in the ablation studies below, so we adopt this method by default. Thereafter, a differentiable renderer $\mathcal{R}$ is utilized to project the modified 3-D model to the background images, and the composite 2-D images can be represented as

$$\mathbf{X}_{\text{in}} = \mathbf{m} \odot \mathcal{R}[(\mathbf{M}_0, \mathbf{T}_0, \mathbf{M}, \mathbf{T}), C] + (1 - \mathbf{m}) \odot \mathbf{X}_0 \tag{1}$$

where $\mathbf{m}$ denotes the object mask obtained during the projection process, $\mathbf{X}_0$ is the background image captured from the virtual simulation environment, and C represents the projection parameters, e.g., camera distance and azimuth elevation angle. The variation of C can result in perturbation regions with diverse shapes.

Given a white-box ViT detector $f(\cdot)$ and loss function $\mathcal{L}(\cdot, \cdot)$, the adversarial patch $\mathbf{T}_{\text{adv}}$ can be optimized according to the following formula:

$$\mathbf{T}_{\text{adv}} = \arg\min_{\mathbf{T}} \mathcal{L}(f_\theta(\mathbf{X}_{\text{in}}), \text{GT}) \tag{2}$$

where GT denotes the object's ground truth. In this work, we are concerned about untargeted attacks and choose the loss function as follows:

$$\mathcal{L} = \mathcal{L}_d + \mathcal{L}_s \tag{3}$$

where $\mathcal{L}_d$ means the detection loss, which is defaulted to the average value of the predicted top-k scores in our approach. This loss term is minimized during the optimization of perturbation patches, resulting in adversarial patterns that can prevent models from recognizing the target. Generally, adversarial patterns with high variance tend to exhibit poor performance in physical attacks. To minimize the variance of the perturbed pixels, we adopt the smooth loss $\mathcal{L}_s$ as follows:

$$\mathcal{L}_{\text{smooth}} = \sum_{e \in E} |e| \cdot |V(F_1) - V(F_2)| \tag{4}$$

where $V(F_1)$ and $V(F_2)$ represent the RGB values of two adjacent triangular facets F_1/F_2 that share the common edge e, with $|e|$ indicating the length of that edge. In addition, E comprises the set of edges within the perturbation mesh, excluding those along the boundaries.

B. Data Augmentation

Data augmentation is known to promote adversarial transferability by alleviating the overfitting problem for CNN-based models. Given that ViT-based models typically split original images into different patches before extracting features, we propose to diversify the corresponding inputs with both global augmentation and local augmentation

$$\widetilde{\mathbf{X}}_{\text{in}} = \text{LA} \circ \text{GA}(\mathbf{X}_{\text{in}}) \tag{5}$$

where the circle notation $\circ$ signifies the composition of mapping functions, representing the combined application of two or more mapping functions in sequence. In addition, $\text{GA}(\cdot)$ denotes global augmentation and $\text{LA}(\cdot)$ represents local

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augmentation. The former applies transformations to the whole image and the latter to certain local patches. Concretely, we adopt GA(·) as the following former in this article:

$$\mathrm{GA}(\cdot) = B_G \circ C_G \circ S_G \circ R_G(\cdot) \tag{6}$$

where B_G represents global brightness, C_G denotes global contrast, $R_G(\cdot)$ hints the global random rotation, and $S_G(\cdot)$ indicates the global random scaling, all of which are conducted with small factors to avoid abnormal scenes. The local augmentation LA(·) can be expressed as follows:

$$\mathrm{LA}(\cdot) = \mathrm{fold} \circ R_L \circ S_L \circ \mathrm{unfold}(\cdot) \tag{7}$$

where unfold(·) unpacks input images into a series of smaller patches, fold(·) represents the corresponding inverse process, R_L selects a certain proportion of unfolded local patches to conduct random rotation, and S_L conducts scaling operations on the selected local patches.

C. Jacobian Matrix Regularization

1) Calculating the Jacobian of Self-Attention: Dot-product self-attention is one of the most critical components for ViTs. The previous work [33] analyzed the total derivative (Jacobian) of self-attention from the perspective of Lipschitz continuous and found that the Lipschitz constant of the standard self-attention was unbounded. Inspired by their work, we provide an insight into the Jacobian of self-attention from another perspective (i.e., adversarial attack) to enhance attack transferability.

Suppose that the self-attention input is $\mathbf{X} \in \mathbb{R}^{N\times D}$, which consists of N tokens

$$\mathbf{X} = \left[\mathbf{x_1}, \quad \cdots \quad, \mathbf{x_N}\right]^\top \tag{8}$$

where each token has an embedded dimension (channel numbers) of D, i.e., $\mathbf{x_i} \in \mathbb{R}^D$. Generally, there are multiple heads in a self-attention module and the output of one head is

$$\mathrm{SA}(\mathbf{X}) = \mathrm{softmax}\left(\frac{\mathbf{X}\mathbf{W}_Q(\mathbf{X}\mathbf{W}_K)^\top}{\sqrt{D/H}}\right)\mathbf{X}\mathbf{W}_V \tag{9}$$

where H is the total number of heads, and $\mathbf{W}_Q$, $\mathbf{W}_K$, and $\mathbf{W}_V$ are learned projection parameters. Afterward, the outputs of all heads are concatenated and then multiplied by another projection matrix. Given that the combination of different heads has no essential effect on the derivative of self-attention, we simplify the output by omitting $\mathbf{W}_V$ and rewriting it as follows:

$$\begin{aligned} g(\mathbf{X}) &= \mathrm{softmax}\left(\mathbf{X}\mathbf{A}^\top\mathbf{X}^\top\right)\mathbf{X} \\ &= \left[g_1(\mathbf{X}) \ \ldots \ g_N(\mathbf{X})\right]^\top \end{aligned} \tag{10}$$

where $\mathbf{A}^\top = \mathbf{W}_Q(\mathbf{W}_K)^\top/(D/H)^{1/2} \in \mathbb{R}^{D\times D}$ and $g(\mathbf{X}) \in \mathbb{R}^{N\times D}$. Since $g(\mathbf{X})$ and $\mathbf{X}$ are both matrices, it is far from straightforward to calculate the derivative of $g(\mathbf{X})$ concerning $\mathbf{X}$. Fortunately, the problem could be simplified by splitting $g(\mathbf{X})$ into vectors of $g_i(\mathbf{X})$. Let $\mathbf{P} = \mathrm{softmax}(\mathbf{X}\mathbf{A}^\top\mathbf{X}^\top) \in \mathbb{R}^{N\times N}$ with scalar elements of $P_{ij} = \mathrm{softmax}(\mathbf{x}_i^\top\mathbf{A}^\top\mathbf{x}_j)$; then, $g_i(\mathbf{X}) = \sum_{j=1}^N P_{ij}\mathbf{x}_j$, and the derivative of $g(\mathbf{X})$ to $\mathbf{X}$ (i.e., Jacobian) can be written as

$$J = \begin{bmatrix} \frac{\partial g_1(\mathbf{X})}{\partial \mathbf{x}_1} & \cdots & \frac{\partial g_1(\mathbf{X})}{\partial \mathbf{x}_N} \\ \vdots & \ddots & \vdots \\ \frac{\partial g_N(\mathbf{X})}{\partial \mathbf{x}_1} & \cdots & \frac{\partial g_N(\mathbf{X})}{\partial \mathbf{x}_N} \end{bmatrix} \tag{11}$$

where $\mathbf{J}_{ij} = \partial g_i(\mathbf{X})/\partial \mathbf{x}_j \in \mathbb{R}^{D\times D}$, which has the following form:

$$\begin{aligned} \mathbf{J}_{ij} &= P_{ij}\left[\left(\mathbf{x}_i\delta_{ij} - \textstyle\sum_{j=1}^N P_{ij}\mathbf{x}_j\right)\right)\mathbf{x}_i^\top\mathbf{A}^\top + \mathbf{I}\Bigg] \\ &\quad + \left[\sum_{k=1}^N P_{ik}\mathbf{x}_k\mathbf{x}_k^\top - \left(\sum_{k=1}^N P_{ik}\mathbf{x}_k^\top\right)\left(\sum_{k=1}^N P_{ik}\mathbf{x}_k\right)\right]\mathbf{A}\delta_{ij} \\ &= P_{ij}\left[\left(\mathbf{x}_i\delta_{ij} - g_i(\mathbf{X})\right)\mathbf{x}_i^\top\mathbf{A}^\top + \mathbf{I}\right] \\ &\quad + \begin{bmatrix} \mathrm{Cov}(\mathbf{X_{:1}}, \mathbf{X_{:1}}) & \cdots & \mathrm{Cov}(\mathbf{X_{:1}}, \mathbf{X_{:D}}) \\ \mathrm{Cov}(\mathbf{X_{:2}}, \mathbf{X_{:1}}) & \cdots & \mathrm{Cov}(\mathbf{X_{:2}}, \mathbf{X_{:D}}) \\ \vdots & \ddots & \vdots \\ \mathrm{Cov}(\mathbf{X_{:D}}, \mathbf{X_{:1}}) & \cdots & \mathrm{Cov}(\mathbf{X_{:D}}, \mathbf{X_{:D}}) \end{bmatrix}\mathbf{A}\delta_{ij} \end{aligned} \tag{12}$$

where $\mathbf{I} \in \mathbb{R}^{D\times D}$ is an identity matrix, δ_{ij} is the Kronecker delta, Cov(:, :) denotes the covariance of two variables, and $\mathbf{X}_{:i}$ represents the sequence of tokens in the view of the ith channel

$$\mathbf{X}_{:i} = [x_{1i}, \ldots, x_{Ni}]^\top \tag{13}$$

where the scalar x_{ki} means the ith value (channel) of $\mathbf{x_k}$.

The Jacobian matrix in (11) contains all first-order partial derivatives of self-attention that influences back-propagation, and the general form of its element consists of two items as shown in the right-hand side of (12). It is obvious that the first item of $\mathbf{J}_{ij}$ has a close relationship with P_{ij} and the second item includes a covariance matrix $\mathbf{C}$, each diagonal of which denotes feature variance in a particular channel and the off-diagonal elements display the covariance of features between different channels.

2) Regularizing the Jacobian Matrix: For a certain model $h : \mathbb{R}^m \to \mathbb{R}^n$ that is Lipschitz continuous, there exists a constant $L = \sup_{x\in\mathbb{R}^m} \|\mathcal{D}_\mathbf{x}h\|$ that satisfies the following condition for any inputs $\mathbf{z}_1, \mathbf{z}_2 \in \mathbb{R}^m$ [34]:

$$\|h(\mathbf{z}_2) - h(\mathbf{z}_1)\| \le L\|\mathbf{z}_2 - \mathbf{z}_1\|. \tag{14}$$

Generally, a smaller L corresponds to a flatter loss landscape and better model generalization [34], which means that the model would be more insensitive to input variation. However, self-attention is not Lipshitz continuous because the second term of $\mathbf{J}_{ij}$ has a close relationship with feature variance, which is input-related and may contain a considerable variance. Nevertheless, we can draw lessons from the above idea by directly constraining the Jacobian when generating transferable adversarial samples with ViT detection models. Unfortunately, it is pretty hard to compute Jacobian matrices in the process of optimizing adversarial samples. To solve the problem, we propose to regularize the subterms of $\mathbf{J}_{ij}$ by removing possible extreme values without directly calculating

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Algorithm 1 Transferable Attack With JMR Method

Require: a subset of layers $\mathfrak{L}$ for the given backbone $\mathfrak{B}$, hyperparameters k and s.
Output: Adversarial texture $\mathbf{T}_{\text{adv}}$.

```
for the number of epochs do
    apply data augmentations based on Equation (5).
    for module m in 𝔏 do                ▷ Forward Propagation
        if m is LayerNorm then          //FVR
            calculate variances Cov(X_:i, X_:i).
            obtain X_:i with top-k variances.
            X_:i ← 0 .
        else if m is self-attention then    //AWR
            obtain the top-k attention weights P_ij.
            P_ij ← 0.
        end if
    end for
    for module m in 𝔅 do                ▷ Backward Propagation
        if m is LayerNorm then
            obtain the gradients G.
            G ← G · s.
        end if
    end for
end for
```

the matrix. Specifically, our methods can be divided into steps shown in Fig. 2(c).

a) Feature variance regularization: The purpose of FVR is to remove the maximum elements of $\mathbf{C}(\mathbf{X})$ in (12). As covariance matrices are positive semidefinite and all principal minors are nonnegative, the main diagonal must contain one element with the maximum absolute value. Therefore, we exclude the off-diagonal covariances and only calculate the diagonal variances. Based on the analysis above, FVR can be achieved with the following steps: 1) calculate the feature variance along spatial dimensions for each channel of a particular layer; 2) locate channels with top-k maximum variances; and 3) reset the features of the selected channels as zeros. The flowchart of the FVR module is demonstrated in Fig. 2(c1).

It should be noted that if we remove the maximum variance of the main diagonal, the covariances in the same row and column are also deleted. In addition, the above method is performed during forward propagation, so we should be aware of which features are suitable to be deleted without bringing in severe impacts to the inference. Considering that mainstream ViT-based detectors usually adopt hierarchical backbones and the number of channels in deep layers is several times larger than that of shallow layers, our variance regularization is mainly carried out in deep layers of backbones. Given that the core of our method is to alter the input variances of self-attention, we choose to perform FVR behind LayerNorm in practice.

b) Attention weight regularization: In addition to regularizing feature variances, we also take into account the influence of self-attention weights, which plays an important role in the first term of (12). To simplify the analysis, we assume that the projected token $g_i(\mathbf{X})$ is insensitive to the variation of attention distributions, so the first term degrades to a linear relationship with P_{ij}. Consequently, we can regularize this term by directly masking top-k maximum attention weights, thereby restricting the potential maximum gradients and mitigating the gradient fluctuations of the model. Fundamentally, this approach aligns with the concept of gradient penalization in optimizing deep learning weights, where a smoother gradient landscape correlates with improved model generalization across diverse data distributions [34]. Similarly, the gradient smoothing operation could enhance the adversarial examples' generalization capabilities across various models.

To further enhance attack transferability, we also apply a decay factor s to the gradients of all LayerNorm modules. The overall method is shown in Algorithm 1.

IV. Experimental Results

A. Experimental Settings

1) DataSet: We follow the dataset of Zhang et al. [10] to train and test perturbation samples. The dataset is collected from CARLA [35] at different distances and camera angles to simulate UAV aerial backgrounds. In addition, each image has a target vehicle to be camouflaged.

2) Detectors Training: To produce adversarial examples, we select four typical ViT-based detectors as white-box models, the backbones of which include Swin Transformer [6], DaViT [7], CrossFormer [36], and plain ViT [37]. Given adversarial samples manufactured on each white-box model, we test their attack performance on all the models mentioned above, as well as two additional detectors with backbones of PVT [38] and MViTv2 [39], respectively. Apart from plain ViT and MViTv2, which are implemented under Mask RCNN prediction head [40], all other backbones are connected to the detection head of Retinanet [41]. In addition, FPN [42] is utilized among all the detectors except ViTDet [37]. For the convenience of description, we simplify ViT-based detectors with corresponding backbones in the following content. To verify the transferable performance toward CNN-based detectors, we choose five models to be attacked, i.e., Mask RCNN [43], SSD [44], Yolo-v3 [45], Yolo-v5,[1] and Yolox [46]. Besides, Detectron2 [47] is selected as the basic framework in our experiments. All detectors are trained on Visdrone2019 [48] and then fine-tuned on the dataset collected from CARLA.

3) Evaluation Metrics: Attack success rate (ASR) is one of the most common criteria for adversarial attack [49], [50], and we take it as our first metric with a confidence threshold of 0.5. Specifically, this article calculates ASR by determining the ratio of $(n_0 - n_1)/n_0$, where n_0 represents the number of detected targets in clean images and n_1 indicates the number of detected targets in adversarial images. Furthermore, another criterion for evaluating attack methods is average precision (AP). It is computed as the weighted mean of precision at specific thresholds, given by $\text{AP} = \sum_n (R_n - R_{n-1}) P_n$, where P_n and R_n represent the precision and recall values at the nth threshold. Intuitively, AP represents the area beneath the precision-recall curve, providing a measure of the balance

[1] https://github.com/ultralytics/yolov5

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between precision and recall across various decision thresholds. In our experiments, we adopt AP@0.5:0.95, i.e., the mean confidence value of APs ranging from 0.5 to 0.95. Besides, we also adopt average recall (AR) to further assess attack performance, with its definition and corresponding attack results presented in the supplementary material.

4) Baseline Methods: As for adversarial attacks on detection models, one common strategy is to reduce the averaged score or object confidence [10], [12]. We refer to these methods as MSM (i.e., mean score method) for brevity and take it as one baseline. In addition, we choose PNA [32] as another baseline method. Although this method was originally tailored for ViT-based classification models, it can be seamlessly applied to ViT-based detectors, delivering competitive attack performance.

5) Implementation Details: Experiments are carried out with the following parameters: maximum epoch of 3, batch size of 2, and Adam optimizer with a learning rate of 0.01. Besides, k is equal to 1 by default, and the scaling factor s is 0.8. In addition, the experiments are conducted on a desktop with an Intel Core i9-10920X CPU and one Nvidia RTX-3090 GPU.

B. Comparison of Transferability Toward ViT-Based Detectors

In this section, we compare the transferability of our method with that of baselines among ViT-based detectors. All the methods are carried out with the same loss function shown as (3), and the only difference is what kind of augmentation would be applied to intermediate features or gradients. To be specific, MSM does not take any intermediate layer augmentation except for input augmentations as (5), which means that adversarial patterns are optimized with the original forward and backward propagations. Besides, PNA ignores residual gradients totally, and we reproduce the method in all multihead self-attention modules. Unlike existing methods, our proposed methodology mainly pays attention to forward propagation. The results of transferability are shown in Table I, where four white-box detectors are displayed in the first column and the generated adversarial samples are tested with detection models in the first row. It should be noted that the names of the detection models are simplified as the corresponding backbones, and detailed information can be found in the section on experiment settings. Besides, a larger ASR or smaller AP indicates better transferability.

According to the second row of Table I, the ASR of our method surpasses that of the majority baselines when transferring from SwinT to other models. Particularly, our method has a clear superiority in terms of ASR when transferring to CrossFormer (51.3%) and MViTv2 (59.5%). Although our method does not have the best APs when transferring to SwinT, Plain ViT, and PVT, the corresponding ASRs are the largest. As for the transferability from DaViT to other models (shown in the third row of Table I), our approach goes almost entirely beyond baseline methods, except a suboptimal AP (65.9% versus 65.7%). The results in the fourth row reveal that our method has advantages over other methods when transferring from CrossFormer to other detectors, but it should be noted that PNA could achieve the best performance when transferring to Plain ViT. In addition, the last row again verifies that our approach could achieve competitive black-box transferability among typical ViT-based detectors, despite the best white-box attack performance realized by MSM. To intuitively recognize the generated patterns, we display typical adversarial samples in Fig. 3, where (a)–(d) denote images generated under our method with SwinT, DaViT, CrossForm, and Plain ViT, respectively.

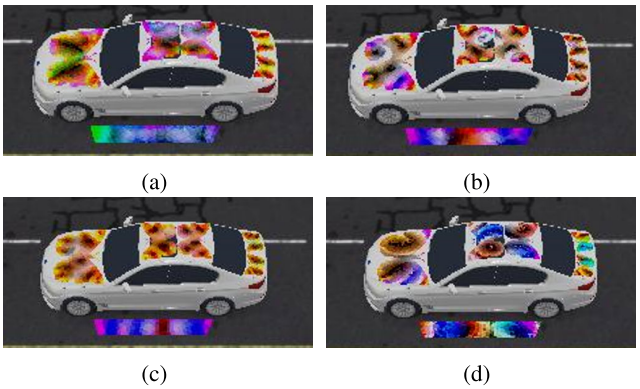


Fig. 3. Typical adversarial patterns. (a)–(d) Samples generated under our attack method with SwinT, DaViT, CrossFormer, and Plain ViT, respectively.

Overall, while our proposed method may lag behind the baseline in certain instances, it consistently outperforms in most cross-model attack experiments. In addition, while PNA distinguishes itself by directly interrupting gradients between patch interactions, its excessive regularization can lead to unintended consequences. For example, as depicted in Table I, when transferring from SwinT, DaViT, and CrossFormer to the target SwinT model, PNA's attack performance is inferior to MSM, which does not rely on gradient operations. In contrast, our approach selectively modulates the Jacobian matrices of self-attention, focusing on terms with the maximum feature variances and attention weights. This selective modulation allows us to achieve more efficient results in the majority of attack scenarios.

C. Attacking Images of Different Distributions

In Section IV-B, adversarial patches are tested on images of a specific distribution. However, it remains uncertain whether these patches would retain their effectiveness in alternative scenarios. To address this query, we conduct further experiments on images exhibiting diverse distributions. Notably, the CARLA dataset provides a specific camera parameter C for each image to ensure accurate rendering of 3-D model targets into 2-D images. This constraint poses a challenge in adopting other datasets when aiming for a natural rendering. To overcome this challenge, we leverage stable diffusion to craft a novel dataset. Fig. 4(d)–(f) depicts typically generated images of different distributions (in rainy scenes), while (a)–(c) showcases the original counterparts. It is worth mentioning that only the backgrounds undergo reconstruction, leaving the foreground targets unaltered. For the details of the generation process, please refer to the Supplementary Material.

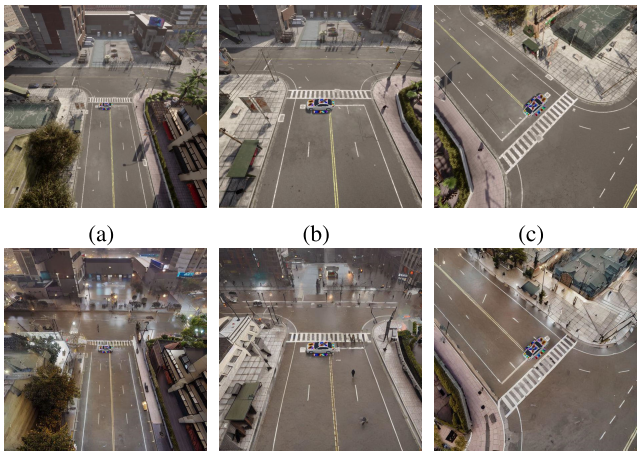
The attack performance is presented in Table II, clearly indicating that our proposed approach maintains superior performance in most scenarios. Specifically, our ASR and AP metrics outperform other methods when transferring from

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TABLE I

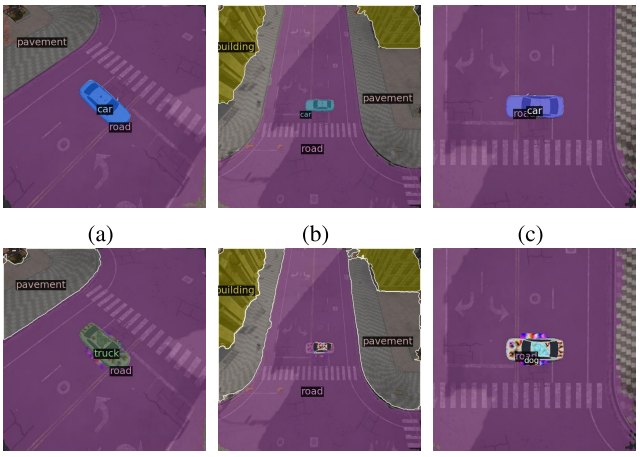
COMPARISON OF ASR AND AP AMONG VIT-BASED DETECTORS WITH DIFFERENT ATTACK METHODS. THE FIRST COLUMN (ROW) DISPLAYS THE BACKBONES OF WHITE (BLACK) BOX DETECTORS. THE BEST ASRS AND APS ARE INDICATED IN BOLD

	Method	SwinT		DaViT		CrossFormer		MViTv2		Plain ViT		PVT	
		ASR(%)	AP(%)	ASR(%)	AP(%)	ASR(%)	AP(%)	ASR(%)	AP(%)	ASR(%)	AP(%)	ASR(%)	AP(%)
SwinT	**MSM**	82.5	**28.1**	45.2	68.7	27.7	61.3	37.2	59.8	4.3	**67.6**	10.2	**66.7**
	PNA	79.5	35.9	59.8	61.5	42.7	56.9	53.3	45.2	4.8	68.7	18.8	69.5
	Ours	**85.0**	31.3	**61.5**	**58.7**	**51.3**	**52.6**	**59.5**	**39.8**	**6.0**	68.8	**18.8**	67.8
DaViT	**MSM**	59.8	51.3	71.5	53.9	52.0	51.8	58.8	39.5	3.2	**65.7**	16.5	**66.2**
	PNA	58.8	53.8	73.2	51.4	55.7	50.5	60.3	37.3	2.7	65.9	18.2	**66.2**
	Ours	**63.3**	**50.2**	**73.7**	**49.3**	**60.5**	**46.8**	**61.5**	**36.9**	**4.2**	65.9	**19.2**	**66.2**
CrossFormer	**MSM**	55.3	**54.7**	53.6	62.9	67.3	38.7	61.0	39.5	6.3	68.5	14.3	68.9
	PNA	54.6	59.3	57.5	59.5	71.2	33.7	**65.5**	33.8	**7.8**	**68.4**	16.3	69.1
	Ours	**59.3**	57.2	**58.8**	**55.9**	**71.5**	**33.4**	65.3	**33.7**	7.0	69.0	**18.7**	**68.8**
Plain ViT	**MSM**	47.6	52.3	36.5	72.0	17.2	66.7	44.3	53.7	**20.0**	**57.0**	12.8	65.7
	PNA	56.6	54.7	47.3	67.9	40.6	57.5	54.0	44.7	15.0	62.3	21.1	66.0
	Ours	**63.7**	**47.2**	**56.2**	**61.3**	**42.3**	**55.6**	**59.2**	40.6	14.5	61.7	**23.5**	**65.1**



(a) (b) (c) (d) (e) (f)

Fig. 4. Typical images generated via stable diffusion. (a)–(c) Original images. (d)–(f) Generated images depicting rainy scenes.



(a) (b) (c) (d) (e) (f)

Fig. 5. Typical segmentation results with SEEM. (a)–(c) Clean images captured from distinct perspectives, (d) and (e) illustrate their corresponding adversarial counterparts.

DaViT or Plain ViT to black-box models. However, there are instances where our method lags behind baseline methods, particularly when transferring from CrossFormer to itself, MViTv2, and Plain ViT. Notably, it is evident that the attack performances in Table II generally outperform those in Table I. This is attributed to the newly generated dataset being outside the detectors' training distribution, thereby diminishing their detection capabilities even in the absence of adversarial patches.

D. Attacks Against SEEM

To further verify the performance of our method, we perform transferable attacks against SEEM [51], one recently proposed interactive segmentation model that could segment everything everywhere all at once. One significant feature of SEEM is that it could output semantic labels to masks in a zero-short manner, allowing us to implement attacks without retraining or fine-tuning the model. To obtain attack transferability, we feed adversarial samples into SEEM with the vision backbone of FocalNet, then count up the number of segmented targets with correct labels, and, finally, calculate ASR with the number of test images.

In the experiment, we evaluate the performance of attack methods with adversarial samples generated by SwinT, DaViT, CrossFormer, and Plain ViT, respectively, and the results are displayed in Table III, where the first column displays the backbones of four white-box detectors and the first row denotes four attack methods. It can be seen that our method could achieve competitive results compared with the baseline approaches. Remarkably, given the samples generated by SwinT, our ASR is about 1.1% higher than PNA and outperforms MSM by more than 16%. Typical segmentation results are shown in Fig. 5, where the first row demonstrates clean image results and the second row denotes the corresponding adversarial images yielded with our attack method. The generated adversarial pattern could prevent SEEM from producing correct labels or masks.

E. Comparison of Transferability Toward CNN-Based Detectors

As there is an excellent difference in exacting features between ViT- and CNN-based models, it is necessary to figure out whether our method could take effect when transferring

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TABLE II
COMPARISON OF ATTACK PERFORMANCE ON THE NEWLY GENERATED DATASET WITH A DISTINCT DISTRIBUTION. THE BEST ASRS AND APS ARE INDICATED IN BOLD

	Method	SwinT		DaViT		CrossFormer		MViTv2		Plain ViT		PVT	
		ASR(%)	AP(%)	ASR(%)	AP(%)	ASR(%)	AP(%)	ASR(%)	AP(%)	ASR(%)	AP(%)	ASR(%)	AP(%)
SwinT	**MSM**	**81.0**	**9.5**	54.0	48.8	38.8	44.4	48.2	37.6	5.0	**53.4**	20.6	49.6
	PNA	79.3	13.8	62.3	37.8	48.8	38.6	61.0	24.7	6.0	54.5	29.2	47.6
	Ours	**81.0**	12.0	**64.0**	**36.3**	**55.2**	**34.6**	**62.3**	**22.9**	**6.2**	53.9	**31.0**	**46.5**
DaViT	**MSM**	66.0	27.4	70.6	27.3	57.5	30.9	65.2	18.9	4.3	**51.2**	31.7	44.6
	PNA	67.0	28.4	72.8	25.6	59.0	29.5	66.3	18.0	5.5	52.1	31.3	45.4
	Ours	**70.3**	**25.3**	**73.2**	**24.2**	**62.3**	**26.7**	**67.5**	**16.7**	**6.5**	**51.2**	**32.0**	**43.7**
CrossFormer	**MSM**	63.5	31.2	57.8	42.1	66.0	21.5	66.1	19.3	6.2	**53.0**	27.0	45.6
	PNA	62.5	33.8	61.0	38.8	**67.3**	**19.8**	**68.6**	**16.5**	6.2	53.9	29.0	45.3
	Ours	**64.5**	**31.2**	**61.2**	**37.3**	65.5	21.0	67.2	17.9	**6.3**	54.4	**30.0**	**44.9**
Plain ViT	**MSM**	55.2	34.4	45.2	53.4	25.5	52.9	54.2	33.0	**17.2**	**36.9**	24.6	47.5
	PNA	65.3	30.8	55.0	46.9	46.0	39.6	63.0	24.0	12.3	42.9	33.8	**42.7**
	Ours	**67.5**	**25.3**	**59.6**	**39.8**	**46.3**	**37.7**	**63.7**	**22.2**	12.5	41.0	**34.5**	**42.7**

TABLE III
COMPARISON OF ASR (%) ON SEEM. THE BEST ASRS ARE INDICATED IN BOLD

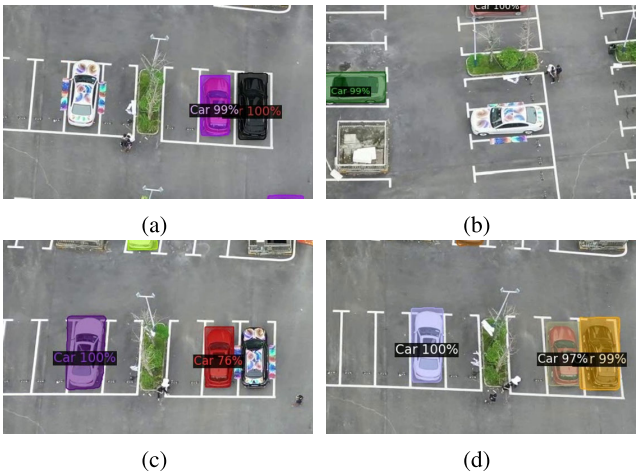
	MSM	PNA	Ours
SwinT	49.3	65.7	**66.8**
DaViT	57.7	57.8	**61.0**
CrossFormer	70.8	71.3	**72.2**
Plain ViT	56.0	62.8	**63.0**

TABLE IV
COMPARISON OF ASR (%) ON CNN-BASED DETECTORS. THE BEST ASRS ARE INDICATED IN BOLD

	Methods	MaskR	SSD	Yolov3	Yolov5	Yolox
SwinT	**MSM**	80.0	80.7	**100**	73.8	93.0
	PNA	90.3	84.7	**100**	87.2	**97.3**
	Ours	**91.0**	**86.8**	99.8	**89.5**	95.5
DaViT	**MSM**	93.5	88.3	**100**	91.7	97.5
	PNA	92.3	**90.8**	**100**	92.5	97.7
	Ours	**93.7**	90.4	**100**	**93.0**	**98.0**
CrossF	**MSM**	92.5	93.7	**100**	86.7	96.5
	PNA	**93.7**	93.2	**100**	**91.2**	**97.3**
	Ours	**93.7**	**94.3**	**100**	89.7	95.5
PlainV	**MSM**	80.8	93.9	99.8	70.0	90.8
	PNA	**89.2**	93.2	**100**	84.5	93.7
	Ours	88.2	**94.1**	**100**	**88.5**	**96.7**

to CNN-based models. To this end, we conduct transferable experiments with several typical CNN-based detectors. The results are displayed in Table IV, where the first column denotes ViT-based source detectors and the first row represents CNN-based target detectors. It should be noted that we abbreviate CrossFormer, Plain ViT, and Mask R-CNN as CrossF, PlainV, and MaskR in the table.

According to Table IV, the majority of ASRs are above 80% for all target detectors adopted, and some ASRs are close to or even equal to 100% (e.g., attacks against Yolo-v3). On the whole, the ASR values appear to be higher than Table I, which implies that adversarial samples generated with ViT-based detectors could be effectively transferred to



(a) (b) (c) (d)

Fig. 6. Typical detection results of physical attacks in UAV remote sensing images. (a) and (b) Depict two viewpoints of a white car equipped with camouflaged patches, (c) illustrates the deployment of the camouflaged patches on a black SUV, and (d) indicates a reference scenario without adversarial patches.

CNN-based detectors, even if samples have moderate or poor attack performance on ViT-based detectors. When comparing different attack methods, our approach outperforms the baselines in most transferable cases except for the penultimate line of Table IV, where PNA takes the lead when transferring from CrossFormer to typical CNN-based detectors.

F. Physical Attacks

To figure out whether our method could make a difference in the physical world, we print the adversarial patterns generated by ViTDet under different attack methods and then deploy the patches on a physical vehicle. Afterward, videos of the target vehicle (a white car in common with the 3-D model mentioned above) are obtained by a miniaircraft UAV at a relative altitude of 60 m and a circle radius of 25 m. Subsequently, the videos captured from different viewing angles are fed into ViT-based detectors to obtain detection outcomes. Typical results are displayed in Fig. 6, where (a) and (b) are two shooting angles of the target vehicle with camouflaged patches, (c) hints that the patches are deployed on a black SUV, and (d) indicates a reference scenario without adversarial patches. The generated patterns could take effect in the physical world.

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TABLE V

COMPARISON OF ASR AND AP FOR ADVERSARIAL SAMPLES WITH DIFFERENT PERTURBATION PATCHES. THE FIRST COLUMN (ROW) DISPLAYS THE BACKBONES OF WHITE (BLACK) BOX DETECTORS. THE BEST ASRS AND APS ARE INDICATED IN BOLD

	Patches	SwinT		DaViT		CrossFormer		MViTv2		Plain ViT		PVT	
		ASR(%)	AP(%)	ASR(%)	AP(%)	ASR(%)	AP(%)	ASR(%)	AP(%)	ASR(%)	AP(%)	ASR(%)	AP(%)
SwinT	**Patch1**	**85.0**	**31.3**	**61.5**	**58.7**	**51.3**	**52.6**	**59.5**	**39.8**	**6.0**	**68.8**	**18.8**	**67.8**
	Patch2	57.8	56.5	27.7	75.6	13.0	73.0	28.8	65.2	0.8	74.7	7.5	78.7
	Patch3	0.0	82.8	3.0	82.8	0.2	77.9	0.0	82.8	0.2	72.6	0.7	76.3
DaViT	**Patch1**	**63.3**	**50.2**	**73.7**	**49.3**	**60.5**	**46.8**	**61.5**	**36.9**	**4.2**	**65.9**	**19.2**	**66.2**
	Patch2	42.0	63.9	34.0	74.2	11.3	72.3	25.0	67.1	0.7	73.2	5.2	77.7
	Patch3	0.0	82.5	9.7	81.6	0.2	76.7	0.8	81.9	0.2	71.2	0.8	74.6
CrossFormer	**Patch1**	**59.3**	**57.2**	**58.8**	**55.9**	**71.5**	**33.4**	**65.3**	**33.7**	**7.0**	**69.0**	**18.7**	**68.8**
	Patch2	25.0	72.8	14.7	78.4	22.0	70.4	37.8	56.8	0.8	74.6	4.8	77.9
	Patch3	0.0	82.9	5.2	83.1	0.2	77.8	0.3	82.7	0.2	72.7	0.8	75.6
Plain ViT	**Patch1**	**63.7**	**47.2**	**56.2**	**61.3**	**42.3**	**55.6**	**59.2**	**40.6**	**14.5**	**61.7**	**23.5**	**65.1**
	Patch2	31.8	68.2	18.1	77.4	5.3	74.7	25.5	66.3	3.5	71.7	4.3	77.9
	Patch3	0.0	82.3	3.2	82.8	0.0	76.9	0.3	82.9	0.0	71.6	1.2	76.0

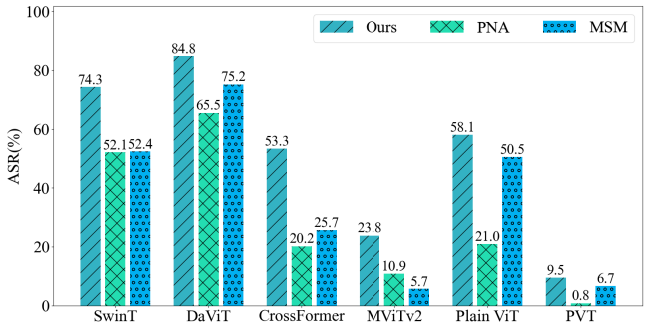


Fig. 7. Comparison of transferability in physical attacks.

To compare attack methods quantitatively, we pick out videos where the target vehicle is camouflaged with different adversarial patches and obtain ASRs by counting the proportion of images with failure detection on the target vehicle. The results are demonstrated in Fig. 7, where the horizontal axis represents the detectors to be attacked and the bars with diverse markings denote the ASRs of different attack methods. According to Fig. 7, our attack method has a clear superiority over other methods on the whole. It should be noted that there are some differences between the ASRs of Fig. 7 and the ASRs in the last row of Table I, which may result from combined effects of complex physical factors, such as illumination conditions, the chromatic aberration of the camera lens, and the relative motion states.

G. Analysis of the Distribution of Feature Variance

As mentioned above, our method applies masks to those features with top channel variances. However, excessive regularization may pose a threat to model prediction, so we need to figure out which stages are suitable for regularization. To this end, we analyze the distribution of feature variances for SwinT layers, which has a hierarchical architecture that is widely adopted by ViT-based detectors. There are four stages in SwinT, and each stage has a certain number of blocks, every one of which has two layer normalization modules (denoted as LN1 and LN2 for convenience), followed by self-attention and MLP, respectively. The distribution of feature variances is displayed in Fig. 8, where the horizontal axis indicates block names of SwinT from shallow to deep layers (e.g., "S2B5" represents the fifth block layer of the second stage), the vertical axis denotes the base-10 logarithm of feature variances, and the boxplots are variance distributions of different channels for features before-LN1, after-LN1, before-LN2, and after-LN2, respectively. Besides, hollow circles are variance outliers of specific channels that are numerically distant from the rest of the channels. According to Fig. 8, despite local fluctuations in feature variance distributions across layers, the maximum variances (i.e., the top outliers) display a general upward trend. Notably, the maximum variances in the latter layers are approximately ten times higher than those in the earlier layers. As discussed earlier, the core of FVR lies in mitigating these peak variances in the Jacobian matrix, ultimately smoothing gradients and boosting attack transferability. Therefore, to achieve better attack performance, our regularization is primarily conducted in deep layers. This will be further validated through ablation studies in Section IV-H.

H. Ablation Studies

1) Effect of Perturbation Regions: As mentioned above, we design a special adversarial mesh that includes two parts of perturbation regions: one part on the vehicle roof and another part on the ground. The reason why we expand perturbation is that the only adversarial patterns on the vehicle could bring about unsatisfactory attack performance. Fortunately, expanding adversarial patterns beyond the object could alleviate the dilemma. To verify the effectiveness of our adversarial patches, we designed two contrastive perturbation meshes based on our original patches by removing the ground mesh and the roof mesh, respectively. The designed patches are displayed in Fig. 9, where (a) implies a clean image, (b) displays the primary patch adopted by this work, and (c) and (d) show the ablation patches. For convenience, we denote the patches in (b)–(d) as "Patch1," "Patch2," and "Patch3," respectively.

The comparison results are displayed in Table V, which has the same layout as Table I except for the second column. Both the ASRs and APs of "Patch1" are much better than those of "Patch2" or "Patch3" for all adopted detectors. For instance, the ASR of the white-box attack on SwinT with "Patch1" is

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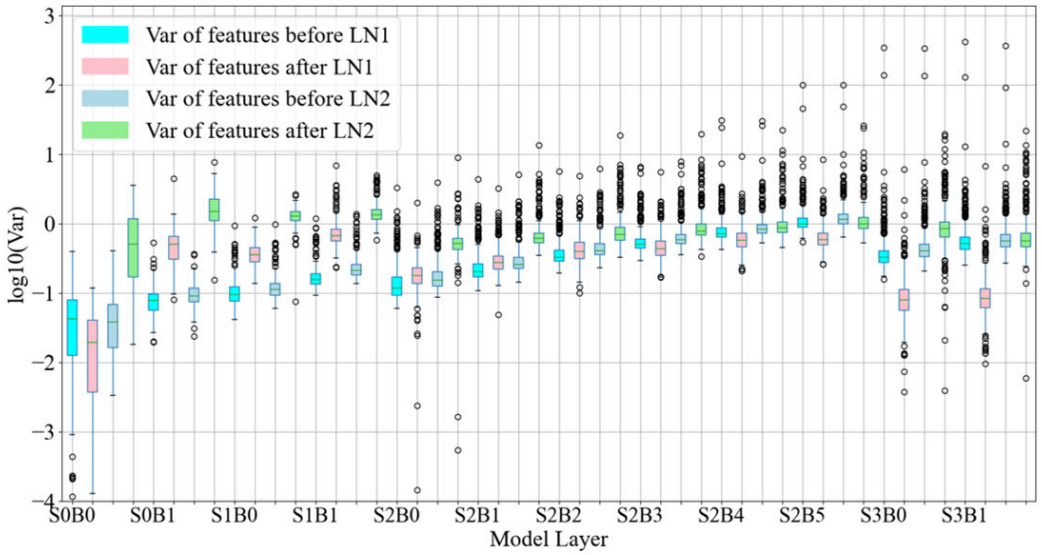


Fig. 8. Feature variance distribution of SwinT.

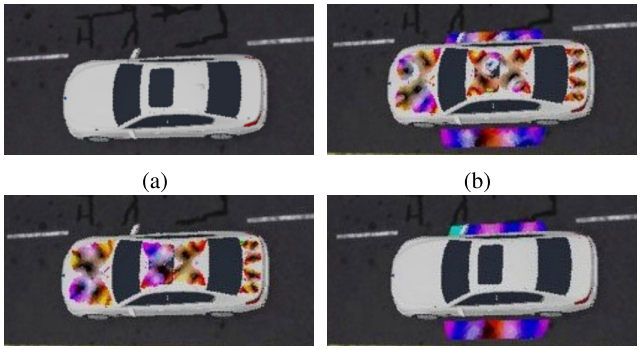


Fig. 9. Adversarial patches with different perturbation regions. (a) Represents a clean image, (b) exhibits the default patch utilized by our method, (c) and (d) illustrate the patches deployed on and off the vehicle, respectively.

85.0%, 27.2% higher than that of "Patch2." The advantage could be further broadened when transferring from SwinT to other models. For example, the ASRs of "Patch1" are more than twice as large as "Patch2" when transferring from SwinT to DaViT, CrossFormer, or MViTv2. Another finding is that the expansion of adversarial patches in a proper manner could take on the effect of "$1 + 1 > 2$," i.e., although "Patch1" is a simple combination of "Patch2" and "Patch3" in visuality, the ASRs of "Patch1" are larger than the sum of ASRs from "Patch2" and "Patch3" individuality. The results reveal that the mere adversarial patterns within or beyond the object just bring about limited transferability, but the combination of them can significantly alleviate the problem. This reveals that ViT-based models are difficult to attack under separate physical patterns, which could be attributed to their excellent ability to capture contextual information.

2) Effect of FVR Level on Transferability: According to (12), the covariance matrix plays an essential role in the Jacobian of self-attention, and the deletion of diagonal outliers could restrain extreme values in the matrix. To understand the influence of FVR, we conduct ablation studies by removing different numbers of feature variances when transferring from ViTDeT to other detectors. Typical results are shown in Table VI, where the first column indicates feature variance settings. For instance, "top-0" means that no FVR is carried out, and "top 1" indicates that we set specific features as zeros for channels with top 1 variances. According to the table, introducing FVR could bring about better overall attack performance. Specifically, the top 2 FVR could enhance obvious improvement on ASR and AP when transferring to DaViT, CrossFormer, and PVT. Besides, top 1 FVR leads to the best transferability for Plain ViT, but the further increase of top-k FVR could produce performance degradation.

3) Effect of AWR Level on Transferability: To verify the influence of AWR, we carry out ablation studies with different AWR settings shown in Table VII, where the first column demonstrates similar settings as in Table VI and the regularization objects are replaced with self-attention weights. It should be noted that attention windows are widely adopted to reduce the computational complexity for ViTs, and thus, our AWR is conducted within spatial attention windows. According to Table VII, compared with top 0 AWR, top 1 AWR can achieve the best performance when attacking SwinT, CrossFormer, and Plain ViT. However, the top 2 AWRs attain the best result for PVT. Overall, although AWR could enhance attack transferability, it plays a less important role than FVR.

4) Influence of FVR Location on Transferability: Mainstream detectors usually employ hierarchical backbones, with each layer concentrating on specific features. Consequently, the choice of which layer to apply FVR can have an impact on attack performance. To assess the influence of FVR location on transferability, we use SwinT as the surrogate model and categorize its backbone into shallow, middle, and deep layers, corresponding to layers 1∼4, 5∼8, and 9∼12, respectively. Following this, we perform top-2 FVR on these three layer sets, and the attack transferability results are presented in Table VIII. According to the table, applying FVR to the deep layers achieves the optimal attack performance for four black-box models. For instance, the ASRs for FVR applied to the deep layers lead the way when targeting DaViT, MViTv2, Plain ViT, and PVT. Therefore, this justifies our decision

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TABLE VI
EFFECT OF FVR LEVEL ON TRANSFERABILITY. THE BEST ASRS AND APS ARE INDICATED IN BOLD

FVR level	SwinT		DaViT		CrossFormer		MViTv2		Plain ViT		PVT	
	ASR(%)	AP(%)	ASR(%)	AP(%)	ASR(%)	AP(%)	ASR(%)	AP(%)	ASR(%)	AP(%)	ASR(%)	AP(%)
top 0	62.7	48.0	54.5	61.8	38.7	58.2	58.3	40.2	13.5	60.9	21.6	65.4
top 1	62.5	**47.5**	54.8	61.4	39.5	56.8	59.7	40.4	**15.8**	**60.7**	22.2	65.7
top 2	**64.0**	47.9	**55.6**	**61.3**	**42.2**	**56.5**	**60.5**	**39.5**	14.0	61.3	**25.1**	**64.6**

TABLE VII
EFFECT OF AWR LEVEL ON TRANSFERABILITY. THE BEST ASRS AND APS ARE INDICATED IN BOLD

AWR level	SwinT		DaViT		CrossFormer		MViTv2		Plain ViT		PVT	
	ASR(%)	AP(%)	ASR(%)	AP(%)	ASR(%)	AP(%)	ASR(%)	AP(%)	ASR(%)	AP(%)	ASR(%)	AP(%)
top 0	62.7	48.0	54.5	61.8	38.7	58.2	58.3	**40.2**	13.5	60.9	21.6	65.4
top 1	**63.0**	**47.3**	54.8	**61.0**	**39.5**	**57.0**	**58.8**	40.4	**16.5**	**60.1**	22.2	65.2
top 2	62.0	48.2	**55.2**	62.5	39.3	58.5	57.5	40.8	15.5	60.2	**22.6**	**64.4**

TABLE VIII
INFLUENCE FVR LOCATION ON TRANSFERABILITY. THE BEST ASRS AND APS ARE INDICATED IN BOLD

FVR location	SwinT		DaViT		CrossFormer		MViTv2		Plain ViT		PVT	
	ASR(%)	AP(%)	ASR(%)	AP(%)	ASR(%)	AP(%)	ASR(%)	AP(%)	ASR(%)	AP(%)	ASR(%)	AP(%)
shadow layers	83.5	31.2	60.8	60.6	**48.2**	53.6	56.0	43.9	4.8	68.3	15.2	68.3
middle layers	**83.8**	**30.0**	54.2	65.1	38.7	57.1	51.3	47.5	3.8	68.7	13.8	**68.1**
deep layers	83.0	32.4	**64.5**	**58.5**	47.0	**52.7**	**57.2**	**43.3**	**6.8**	**68.1**	**15.7**	69.9

TABLE IX
INFLUENCE AWR LOCATION ON TRANSFERABILITY. THE BEST ASRS AND APS ARE INDICATED IN BOLD

AWR location	SwinT		DaViT		CrossFormer		MViTv2		Plain ViT		PVT	
	ASR(%)	AP(%)	ASR(%)	AP(%)	ASR(%)	AP(%)	ASR(%)	AP(%)	ASR(%)	AP(%)	ASR(%)	AP(%)
shadow layers	**86.0**	30.2	61.2	61.4	45.3	54.2	54.2	46.2	5.0	68.3	16.5	69.1
middle layers	83.5	32.5	63.5	**59.2**	**46.8**	**52.3**	**55.7**	**44.2**	**6.5**	**67.9**	**17.5**	68.8
deep layers	83.3	**29.9**	**63.8**	59.6	45.7	53.3	54.0	44.8	6.3	68.2	16.8	**68.6**

to perform FVR on the deep layers in the aforementioned experiments.

5) Influence of AWR Location on Transferability: Finally, we investigate the impact of AWR's application location on attack performance. We use SwinT as an illustrative example and partition its backbone into three parts: shadow, middle, and deep layers. We then apply AWR by masking the top two maximum attention weights in each of these layers and record the attack outcomes in Table IX. The table indicates that for most black-box models, excluding SwinT, applying AWR to the middle layers yields superior results. Although the ASR for middle layers falls slightly behind deep layers when targeting DaViT, the margin is negligible. A similar trend is observed for AP when targeting PVT. In summary, the middle layers tend to be the optimal choice for our proposed AWR method.

I. Discussion

In this section, we delve into how and why JMR can enhance the transferability of attacks against ViT-based detectors. Owing to the disparities in architecture and complexity across various models, adversarial patches tend to suffer a decline in their effectiveness when transferring from surrogate models to black-box counterparts. To enhance attack transferability, many works employ gradient regularization, assuming models exhibit Lipschitz continuity [28], [52]. Nevertheless, this assumption does not apply to typical ViT-based models due to the unbounded nature of the covariance matrix elements in (12). Rather than imposing regularization on the gradients of ViT-based models, we constrain the Jacobian matrix indirectly during forward propagation. Concretely, we propose a method that locates the most significant feature variances and zeros out the corresponding features, thereby controlling the second term of (12). Furthermore, we also modulate the first term by masking the highest attention weights. It is noteworthy that, despite the absence of direct gradient operations in our proposed FVR and AWR, they can influence the gradient landscape through (12). In essence, the operations can limit the extreme value of gradients, i.e., restricting L in (14), regardless of its input dependence for self-attention modules. By employing the approach, we not only mitigate gradient fluctuations but also diminish the excessive focus of the attention module on input data, thereby enhancing the transferability of adversarial examples across various models.

V. CONCLUSION

In this work, we present a method of promoting physical attacks against ViT-based detectors in UAV remote sensing images. Instead of adopting the frequently used setup that requires adversarial patterns to be aligned with image tokens, we propose to optimize adversarial patterns on separate

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meshes deployed within and beyond the target object, which could strike a balance between physical implementation and attack performance. Our results show that only adversarial perturbations within the target object could achieve limited attack performance on ViT-based detectors, but extending perturbations beyond the object could alleviate the problem. In addition, we augment input images at global and local levels to narrow the gap between limited digital space and complex physical space. To further promote attack transferability, we put forward the JMR method that is implemented during forward propagation. Considering the difficulty of calculating Jacobian directly, we divide the regularization process into FVR and AWR, where the former regularizes the features of channels with top-k variances and the latter resets the maximum self-attention weights. Extensive experiments on UAV images are conducted to verify the effectiveness of our method in promoting attack transferability.

As for limitations, the Jacobian matrix of our method is deduced with simple self-attention modules but may not apply to complex attention structures. In addition, while adding adversarial patterns beyond the object could enhance attack performance, this setup limits the application scenarios with static targets in the physical world. Thus, a more powerful and flexible adversarial patch is yet to be explored in future work.

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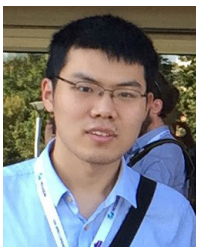
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